

DURHAM, Australia

WMO: 954870

Lat: 27.401S

Lon: 141.811E

Elev: 378

StdP: 14.50

Time Zone: 10.00 (E10)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 41.4	(c) 43.5	(d) 10.4	(e) 9.5	(f) 79.0	(g) 14.4	(h) 11.6	(i) 77.7	(j) 23.4	(k) 69.9	(l) 20.9	(m) 66.5	(n) 8.9	(o) 140	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	108.8	68.0	106.4	67.6	103.8	67.5	76.4	88.6	75.0	88.1	73.5	88.8	11.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	128.0	80.4	71.5	118.0	80.2	69.0	107.9	81.2	40.2	88.5	38.8	88.6	37.3	88.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.8	(b) 22.1	(c) 20.0	(e) 37.3	(f) 112.5	(g) 2.2	(h) 2.9	(i) 35.7	(j) 114.5	(k) 34.4	(l) 116.2	(m) 33.1	(n) 117.9	(o) 31.5	(p) 120.0		
			DB	WB	33.0	77.8	2.0	1.8	31.6	79.1	30.5	80.2	29.4	81.2	28.0	82.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.0	90.1	87.3	82.6	74.7	64.7	57.8	57.4	60.9	68.5	76.2	81.6	86.9
	DBStd	13.13	6.21	6.56	6.35	6.43	5.88	5.44	5.37	6.29	7.26	7.56	7.36	6.69
	HDD50	4	0	0	0	0	0	2	2	0	0	0	0	0
	HDD65	769	0	0	0	6	79	226	244	161	44	8	1	0
	CDD50	8762	1244	1044	1011	741	457	237	232	338	554	812	948	1144
	CDD65	4054	779	624	546	297	71	12	9	35	148	355	499	679
	CDH74	57488	12501	9421	7311	3196	560	65	84	426	1827	4736	6950	10411
	CDH80	34399	8442	6006	4148	1405	104	4	8	121	783	2527	4122	6729
Wind	WSAvg	11.3	12.4	12.3	11.9	10.9	10.1	10.1	9.8	10.7	11.3	12.2	12.1	12.4
Precipitation	PrecAvg	8.3	1.4	1.3	0.9	0.6	0.6	0.4	0.5	0.3	0.5	0.4	0.9	0.7
	PrecMax	27.7	5.3	8.3	4.8	3.1	3.7	2.0	3.5	1.3	2.5	2.4	4.5	3.9
	PrecMin	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.2	1.5	1.7	1.2	0.9	0.9	0.6	0.9	0.4	0.7	0.5	1.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	112.1	110.9	106.3	96.8	86.5	79.8	80.9	89.7	97.8	102.3	109.5	110.8
		MCWB	68.8	69.8	68.2	66.6	61.4	60.4	57.8	60.7	61.7	64.2	67.2	66.1
	2%	DB	109.1	107.1	101.9	93.6	82.7	75.5	76.1	83.1	92.5	98.9	104.8	107.4
		MCWB	68.4	68.2	66.9	64.9	60.7	58.8	56.3	57.0	59.8	62.6	65.7	66.3
	5%	DB	106.4	104.1	98.7	90.9	80.1	71.8	72.7	78.5	88.3	96.0	100.9	104.3
		MCWB	68.5	68.0	66.0	63.7	59.6	56.6	54.7	54.9	57.9	62.2	64.1	65.9
	10%	DB	103.6	101.0	95.9	88.1	76.9	68.4	69.1	74.5	84.3	92.9	97.2	101.4
		MCWB	68.8	68.0	66.0	62.2	58.1	55.2	53.4	53.6	57.0	61.3	63.7	66.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.5	77.9	76.2	72.9	67.4	64.3	66.0	63.0	67.1	69.4	72.6	75.9
		MCDB	90.1	89.0	80.9	83.0	76.4	71.6	71.0	77.9	75.8	85.3	86.9	87.8
	2%	WB	76.6	76.3	74.3	69.4	64.0	62.0	60.7	59.9	63.5	67.6	70.8	73.4
		MCDB	88.1	91.1	83.1	85.1	75.8	70.4	68.7	76.2	80.4	85.9	86.7	89.7
	5%	WB	75.1	75.0	72.7	67.2	61.8	59.6	57.3	57.0	61.3	65.4	69.5	71.4
		MCDB	88.7	90.0	85.4	84.1	76.2	68.1	68.5	76.8	81.4	87.4	86.3	90.0
	10%	WB	73.5	73.1	70.9	64.9	59.7	57.1	54.4	54.8	59.0	63.2	68.0	69.9
		MCDB	91.2	88.1	86.0	82.8	74.0	66.1	67.1	72.4	82.1	88.1	86.7	91.5
Mean Daily Temperature Range	5% DB	MDBR	21.8	21.1	21.1	21.6	21.0	19.5	21.1	23.3	24.1	25.0	23.5	22.6
		MCDBR	23.5	23.8	23.8	23.3	23.0	22.4	24.6	27.4	29.1	27.5	27.5	25.2
		MCWBR	7.3	7.5	9.0	9.2	10.5	11.1	11.9	12.8	12.2	11.1	9.9	9.1
	5% WB	MCDBR	18.9	17.4	17.4	19.4	19.4	17.7	20.5	24.4	22.5	23.6	21.6	20.4
		MCWBR	6.2	6.3	7.3	8.0	9.9	9.8	11.1	12.2	11.2	11.2	7.7	8.8
Clear-Sky Solar Irradiance	taub		0.365	0.359	0.342	0.325	0.308	0.297	0.299	0.314	0.361	0.359	0.368	0.364
	taud		2.440	2.481	2.505	2.513	2.510	2.535	2.511	2.445	2.321	2.348	2.379	2.411
	Ebn at Noon		310	306	302	293	284	280	284	292	291	303	307	311
	Edh at Noon		39	36	34	30	27	25	27	32	40	41	41	40
All-Sky Solar Radiation		RadAvg	2498	2338	2052	1701	1328	1143	1253	1565	1936	2295	2414	2509
		RadStd	103	153	128	91	61	66	65	75	119	120	147	135

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	+1.80	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page